Marut Mahija Roy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.82
INDIAN PUBLIC SCHOOL	Class XII (CBSE)	2018	75.60%
ST. XAVIER'S ENGLISH MEDIUM SCHOOL	Class X (CBSE)	2016	9.40%

TECHNICAL SKILLS

Programming: Python, Verilog HDL

EDA & Circuit Design: Cadence, KLayout, LTspice

Data Analysis & Visualization: NumPy, Pandas, SciPy, SymPy, Matplotlib, Plotly, scikit-learn

Tools & Platforms: GitHub, Jupyter Notebook

Thin-Film Deposition & Processing: RF/DC magnetron sputtering (PVD), ion milling, thermal annealing, high-temperature deposition

Electrical & Magnetotransport: Two-probe, four-probe, probe station, SMU, I-V characterization, impedance analyzer, Seebeck, Hall, AHE

Magnetic Characterization: MOKE, FMR, VSM

Structural & Material Characterization: Powder XRD, Thin-Film XRD, Raman, UV-Vis, SEM, TEM

RELEVANT COURSES

Semiconductor Devices, Fundamentals of Semiconductor Materials, Micro and Nanofabrication, Advanced Nanofabrication and Characterization Techniques for Next Generation Devices, Semiconductor Devices Characterization Lab, Semiconductor Materials Characterization Lab, Vacuum Technology in Semiconductor, Fundamentals of Electronic Packaging, CMOS Digital Design, Diffraction and Spectroscopy, Fundamentals of Scientific Computing, Imaging in Materials Science

EXPERIENCE

IoT Intern - Technophilia Solutions

In association with Microsoft Business Club (B-Club), ***IIT Kharagpur*** | **Aug 2022 - Sep 2022**

- Completed an **IoT-focused internship program** involving sensor interfacing, microcontroller programming, and embedded-system development.
- Designed and simulated an **automated Smart Dustbin** using **Arduino Uno, ultrasonic sensor, servo motor, and PCB circuitry** in **Tinkercad**.
- Programmed the Arduino Uno using **Arduino IDE** to interface the ultrasonic sensor and control **servo-actuated lid operation**.

M.TECH THESIS / RESEARCH EXPERIENCE

Investigation of Structural and Magnetic Properties of High-Entropy Alloys for Spintronic Magnetic Device Applications

Guide: Prof. Pinaki Prasad Bhattacharjee | **Co-Guide:** Dr. Chandrasekhar Murapaka

- Investigated **AlCoCrFeNi2.1 HEA thin films (100 nm)** deposited on **p-type Si** using PVD sputtering, with post-deposition annealing and high-temperature deposition at **400 °C**.

- Performed **XRD, EBSD, SEM, MOKE, and FMR** characterization, along with **resistivity and AHE measurements**, to evaluate structural, magnetic, and transport properties.
- Observed **lower switching field and softer magnetic behavior** in high-temperature-deposited films compared with annealed films, with greater sensitivity to electrical/magnetic noise during switching.
- **Planned work:** Investigate deposition at **300-600 °C** to identify optimal processing conditions, followed by **Spin Hall Effect measurements using a different alloy system** and magnetic device fabrication.

SCHOLASTIC ACHIEVEMENTS

- **University Gold Medalist - ECE (2023):** Awarded the Gold Medal for securing **First Position in the Electronics & Communication Engineering batch**.
- **JEE Main - 2019:** Qualified JEE Main and secured admission to the **B.Tech program at Assam University**, a Central University and GFTI.

POSITIONS OF RESPONSIBILITY

Placement Coordinator - SMD M.Tech, Department of MSME, IIT Hyderabad

- Coordinating **placement-related activities**, communicating with the **Office of Career Services (OCS)**, and serving as a liaison between students and recruiting organizations.

OCS Volunteer - Office of Career Services, IIT Hyderabad

- Supported **placement-related activities and coordination** as an OCS volunteer during the first year of M.Tech.

Placement Coordinator - Electronics & Communication Engineering, Assam University

- Coordinated **department-level placement activities** and facilitated communication between students and the placement team during the final year of B.Tech.

CERTIFICATIONS & TRAINING

Three-Day Job-Oriented Training Programme for Final Year Engineering Students

Career Counselling and Placement Cell, Assam University | **Nov. 28-30, 2022**

AREAS OF INTEREST

Semiconductor Devices, Semiconductor Materials & Thin Films, Semiconductor Process Technology, Semiconductor Packaging, Nanofabrication, VLSI & CMOS Design, Magnetic Materials & Spintronics, Magnetotransport, Magnetic Device Applications

PERSONAL INTERESTS

Sports: Cricket, Football, Badminton | **Creative:** Painting | **Leisure:** Travelling, Music